Atomisation

Breaking learning into prerequisite and new knowledge

Use atomisation in lesson preparation to identify the small pieces of knowledge and action a successful performance requires. Separate existing prerequisites from genuinely new learning.

How to do it well

1 Choose the endpoint. Write a representative question, performance or response that embodies the intended learning.

2 Unpack the actions. Work through it and list the knowledge, interpretations and decisions required at each point.

3 Separate old and new. Mark components that students should already know and those the lesson must explicitly teach.

4 Plan the prerequisite checks. Use short direct prompts, repair insecure foundations and recheck before introducing dependent content.

5 Build the episode. Sequence the new components, a concise model, guided rehearsal and recombination into an independent whole-task attempt.

What this sounds like

“To calculate this mass, students must interpret the formula, find the amount, use the relationship and manage units. Which are prerequisites, and which are new today?”

Classroom example. Preparing a gas-volume calculation, the teacher separates interpreting the question, converting units, recalling the relationship and using the amount correctly. A short check reveals that unit conversion is insecure. That prerequisite is repaired before the new application is modelled. Atomisation changes the teaching route rather than merely producing a longer planning list.

Top tips that make the technique work

Start with a real task. A representative example exposes the decisions students actually need to make.

Ask what each step depends on. Reading a symbol, selecting a rule and completing an operation may require different knowledge.

Check prerequisites separately. A complex final question can hide which small component is missing.

Avoid endless fragmentation. Keep the overall concept visible and recombine the pieces after rehearsal.

Use the analysis to plan teaching. Atomisation is useful when it changes the explanation, check or practice sequence.

Do and avoid

Do	Avoid
Identify prerequisite and new components explicitly.	Reducing the task to disconnected facts.
Plan a direct check for the likely bottleneck.	Assuming a difficult final answer reveals which component failed.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Atomisation